

JACOB WU-CHEN

Boston, MA | Washington, DC | Seattle, WA | Available July 2026

📞 301-385-4159

✉️ wu-chen.j@northeastern.edu

🌐 jakewc.com

🌐 linkedin.com/in/jacob-wu-chen



github.com/jakewc12

Education

Northeastern University

Sep 2022 – May 2026

B.S. in Computer Science, Concentration in Artificial Intelligence

Boston, MA

- GPA: 3.8/4.0 | Dean's List
- Activities: Student Government Association, Electric Racing Club, Club Golf
- Relevant Coursework: Algorithms, Computer Systems, Object-Oriented Design, Machine Learning & Data Mining, Natural Language Processing

Technical Skills

Languages: Python, TypeScript, Java, Kotlin, SQL, C, HTML/CSS

Frameworks & Libraries: React, Next.js, FastAPI, Flask, PyTorch, NumPy, Playwright, Vitest, PyTest

Tools & Platforms: Git/GitHub, AWS (EC2, Lambda), Docker, VS Code, IntelliJ, Conda, Kali Linux

Work Experience

MORSE Corp

Jul 2025 – Dec 2025

Full-Stack Engineer Co-op

Cambridge, MA

- Developed and extended FastAPI endpoints, supporting new full-stack features and improved API usability
- Abstracted data migration configurations to allow web services to use both locally hosted and containerized databases
- Increased frontend unit test coverage by **6×** and authored frontend testing standards for an entire code repo
- Added **over 5** new frontend features and pushed frontend fixes for **over 10** feature bugs
- Created a shell script to **automate docker production deployment**, removing user dependency during deployment

Air Force Research Laboratory

Aug 2024 – Dec 2024

Machine Learning & Computer Vision Co-op

Rome, NY

- Developed a detection model to separate in-distribution and out-of-distribution data with an **86 %** accuracy rate utilizing Convolutional Neural Nets and Vision Transformers
- Researched state-of-the-art image segmentation and classification models to facilitate the understanding and development of personal-use ML models
- Produced clear and concise documentation on a variety of machine learning papers involving computer vision, image segmentation, and object detection

Northeastern University Student Government Association

Jan 2024 – Present

Tech Lead & Software Developer

Boston, MA

- Led a **team of 5 engineers** to build a senator nomination platform **with 300+ active users**
- Designed and implemented a production level full-stack app using TypeScript, React, HTML, CSS, NestJS, and a Supabase backend, then deployed the platform with Docker pipelines to ensure functionality and scalability across environments

Projects

Senate Attendance Manager

Jun 2025 – Present

TypeScript, Next.js, Supabase, Vercel

- Created a web app to allow NEU SGA Senate to take roll-call for meetings, ensuring accurate attendance records
- Maintained and updated a production environment with **50 active users**, as well as dev and staging environments
- Created a CI/CD pipeline, automating production deployments, running database migrations, integration tests
- Designed and developed a full-stack application using a NextJS tech suite, allowing for seamless and serverless communication between the frontend and backend

Health Disease Detector

May 2024 – Jul 2024

Python, React, Flask

- Achieved **95 % precision and accuracy** with trained Machine-Learning models to predict the likelihood someone possesses heart disease or diabetes based on personal health info
- Created smooth-loading and seamless web pages by designing an efficient RESTful application, using React web pages to connect to Flask endpoints

Interests

Golf, Indoor Rock Climbing, Piano, Cooking, Traveling, Robotics